Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	
Project Name	FORMULA 1 Race Intelligence Dashboard Analysis Report (2010-2026)
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect to the supplied Excel workbook and use race_results, circuits, championship_standings, lap_intelligence and qualifying_results.	Medium
2	Data Processing	Check data types, nulls, duplicate keys and field consistency; relate tables at correct grain; avoid duplicating rows through joins.	Medium
3	Business Analysis	Calculate points, wins/podium counts, DNF counts/rates, positions gained, tyre usage, lap-time and pit-stop summaries.	Medium
4	Business Insights & Analytical Outputs	Provide driver leaderboard, constructor season trend, tyre compound analysis, host circuit/country view, podium analysis and reliability analysis.	Medium
5	User Interaction & Analysis	Allow filtering by season, team, driver, country/circuit, circuit type, weather/wet race and tyre compound where applicable.	Medium
6	Data Update & Reporting	Show refresh/source date and document assumptions; update by replacing or refreshing the source workbook.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Labels, legends and tooltips must be clear; dashboards should use consistent F1-inspired colours and readable typography.
NFR-2	**Security**	Use only the supplied dataset; avoid exposing credentials or private connection details in published workbooks.
NFR-3	**Reliability**	Use consistent field definitions and verify totals against the source workbook; clearly mark incomplete/unknown values.
NFR-4	**Performance**	Use extracts/aggregations and avoid unnecessary high-cardinality marks; test filters on the full dataset.
NFR-5	**Availability**	Workbook should open and refresh in the intended Tableau environment; publication depends on account/access.
NFR-6	**Scalability**	Design calculations and data relationships so additional seasons or updated race files can be added with minimal redesign.

Project-specific Documentation

Dataset workbook: F1 Racing Project Dataset(1).xlsx.

Sheets and data rows (excluding headers): race_results 6,464; circuits 29; championship_standings 311; lap_intelligence 3,814; qualifying_results 3,130.

Observed race_results coverage: seasons 2010–2026; 334 distinct race_id values; 19 distinct driver labels; 10 distinct team labels; 21 distinct country labels.

Available analysis fields include race date/season/round, circuit type and location, driver/team, finish/grid position, points, DNF and DNF reason, weather/wet-race flag, safety-car/VSC/red-flag data, pit stops, tyre strategy/starting compound, overtakes and laps led.

Lap-level fields include lap/sector times, tyre compound and tyre age, DRS, speed trap and pit-lap flag. Qualifying fields include qualifying position, Q1/Q2/Q3 times, gap to pole and top speed.

Validation note: workbook is a multi-sheet analytical dataset, not necessarily official complete historical timing. Verify values and data types before presenting findings. Do not infer driver nationality: no nationality field is present.

Use COUNTD([race_id]) for race-event counts where appropriate; race_results contains multiple driver-result rows per event. Distinguish race-result row counts from unique Grand Prix counts.

• Recommended analysis: driver points; constructor season trajectory; tyre compound usage; tyre age versus lap time; podium distribution; country/circuit comparisons; DNF by team/season/reason; wet-race/circuit-type reliability; grid-to-finish position gain; pit-stop and safety-car context.

• Limitations: no driver nationality field; DNF reasons are sparse; dataset's historical coverage and synthetic-looking labels should be validated against an authoritative source before external claims.